Lesson Notes

MODULE 2 | LESSON 2

TYPES OF STOCKS AND CRYPTOCURRENCIES

Reading Time	60 minutes
Prior Knowledge	Interest, Shorting, Financing
Keywords	Capitalization, Dilution, Private equity, Price dynamics, Alternative investments, Commodities, Art (a investment)

In the previous lesson, we laid the foundation for what equities and cryptocurrencies are and how their expected return ar volatility can be measured. In this lesson, we'll examine different ways of categorizing these securities, as well as compar these securities to each other and to alternative investments.

1. Large Cap vs. Small Cap

To understand capitalization, first, let's understand what a company is worth. This very simple equation is based on the m capitalization of a firm. The equation is:

$$\textbf{Market capitalization} = \textbf{Number of Shares outstanding} \times \textbf{Current Share Price.}$$

The right-hand side consists of two terms. The first is the number of shares the company has outstanding. Typically, thes shares are created at the time of the IPO. However, a company may issue additional shares. When shares are repurchase company itself, these are known as **treasury stock**. In total, all of the shares combined should represent 100% of the own of the firm. Be sure you don't confuse these with other types of securities, such as bonds or options, which do not pertain ownership. Stock ownership represents an exclusive and complete ownership of the firm.

Typically, the number of shares does not change often over time, but occasionally, it does. Suppose that company A decid buy company B. Company A offers each shareholder of B 1.2 shares of A for each share they own, if they accept the merg requires that company A issues a certain number of new shares or uses a combination of treasury stocks and new shares

When new shares are issued, this dilutes existing investors. For example, if a firm had issued a million shares and decides issue a million more, then they would be **diluting** the original investors, watering down the amount of ownership those ow have. For this reason, when a company needs to raise additional funds and issue new shares, the existing shareholders ty have the right to maintain their proportion of ownership (antidilution provisions) but this has a cost. However, a discussio dilution is beyond the scope of our discussion.

While the number of shares remains relatively fixed, the share price changes daily. In fact, it changes intraday. The share p simply the prevailing price in the market. Generally, this is easy to determine because exchanges publish prices and volum information throughout the trading day. Suppose you were a big investor trying to buy a significant portion of the company takeover). You would need at least the number of shares times the current share price to purchase the company. Of cours price will increase as existing stockholders recognize your ambition to buy their stock and now demand a premium for it, perhaps 30% or more. In addition, one would need to add the net debt as well.

In some instances, companies can decide to do a **share split** to make stocks more affordable (or even vice versa, in what called a **reverse split**). Suppose a company has 1,000,000 shares at $40 per share. This is worth $40 million. The compan decides to do a 2-for-1 split. Now, the new number of shares is 2,000,000 and the share price is $20. The market cap rema same, but this action helps to make the stock more affordable. For example, a round-lot order of 100 shares now costs $2 instead of $4,000. Splits have no effect on market cap, but they can help facilitate easier trading of the stock.

Now that we have defined market capitalization, we can categorize companies into 4 groups:

1. Large cap: $10 billion and up
2. Mid-cap: $2 billion through $10 billion
3. Small cap: $0.3 billion thru $2 billion
4. Micro-cap: up to $300 million ($0.3 billion)

Let's focus on large caps versus small caps. Large caps are clearly more established companies. Sometimes, these are k as blue chips given their large size. (The term blue chip comes from the most expensive chip in poker—blue chips). These companies tend to be the ones that have been around longer and that do well in good economies as well as poor ones. W some examples of large cap companies? These include many companies that are household names throughout the worlc Microsoft, Apple, Amazon, Alphabet, and Meta (formerly Facebook) to name a few in technology, as well as older ones su Visa, Mastercard, and Johnson & Johnson. In 2018, Apple was the first company to ever achieve a market capitalization o trillion.

Small cap companies, on the other hand, tend to be relatively newer companies. While there may be companies that have around for years, most companies will grow to over $2 billion dollars, especially if they are global brands. When Apple laur its IPO in 1980, its market capitalization was $1.8 billion. (At the time, it would have been considered mid-cap, as the thres was lower in 1980 than it is today). Market capitalization is a very clear way to distinguish these companies. Indeed, wher shopping for stocks, one way is to determine what percent goes in each category of large cap and small cap. Due to the la size, large cap stocks are thought to be considered relatively safer than small-cap stocks, given that their size often come experienced management, existing revenue lines, and market share.

2. Growth Stocks vs. Income Stocks

We have already seen the distinction between value/income stocks and growth stocks. Growth stocks are usually those o companies that are in a phase whereby any income the company makes is used as retained earnings to grow the firm's competitiveness. This growth could be in existing markets by improving existing products, bringing new ones to market, expanding the company's geographical reach, or perhaps even expanding into different product markets.

Companies that are in the growth category are clearly identified as such. It is unlikely that an investor who wants dividend select growth stocks. Indeed, without a stream of dividends, it may be difficult to determine if a growth stock is properly v Growth stocks are meant to be long-term investments that will have big capital gains. Depending on the complexity and competitiveness in the industry in which they operate, growth companies can take many years to enter the phase where tl have established a serious market share with a consistent revenue. For example, Microsoft had its IPO in 1986 but did no dividend until 2003. From 1986 through 2003, Microsoft was considered a growth company. In 2003, it began paying divid At that point, you can consider Microsoft an income stock. As you can see, companies can change as they evolve in their industry.

An income stock is simply one that is expected to provide income in the form of dividends. These companies are clearly distinguished from growth stocks because they will have a historical dividend yield. When determining whether to buy an stock, one looks at the recent history of dividend returns. For example, a $200 stock paying a $10 dividend per year would considered to have a 5% dividend yield. Typically, dividends are paid quarterly based on the earnings as well as the earnin peers in the industry. If a dividend stock skipped paying a dividend, the response in the market would be swift and potenti extreme: namely, a sharp decrease in the stock price and negative sentiment.

When comparing stocks to bonds, income stocks clearly look more like "fixed-income securities" (bonds). Unlike bonds th stocks do not guarantee a dividend amount or date. It is possible that stocks may miss a dividend or pay less than expect The word "fixed" in bonds really emphasizes the guaranteed (barring a default) payout.

3. Domestic vs. Foreign Stocks

Stocks can be domestic or foreign, depending on the relationship of the company and the investor. Stocks from companie domiciled in the same country as the investor are **domestic stocks**. Stocks from other countries are **foreign stocks**. A dor stock is one in which the investor buys the company's stock in their own country. It is true that these companies may oper multiple nations worldwide. The key is that domestic stocks are in the same economy and same currency in which the inv does business. It is for this reason that it can be prudent to own foreign stocks as well. The reason holding foreign stocks is that it achieves exposure to other economies, in other markets, and in other currencies. For example, suppose you own domestic stock of a manufacturing company. Now, manufacturing stocks drop. Your domestic stock is bound to decrease

Suppose you hold a foreign stock of a manufacturing company. In that part of the world, perhaps manufacturing stocks d go down. You could actually make money there. Supply and demand can be very local at regional and national levels, so it possible to lose money in a sector in one part of the world and make money in the same sector in another part of the worl international economy is vast and complex. Therefore, it makes sense, from a balanced point of view, to hold some dome stock and some foreign stock.

Moreover, perhaps the exchange rates are favorable. Consider the same example where the foreign manufacturing stock remains unchanged. However, suppose the exchange rate in that country improves relative to your country. That means th foreign currency is stronger. When you go to sell your foreign stock, you will get that currency, which you can then convert greater number of units of your local currency. For example, if the currency in your own country loses 10% relative to the o country but the return is the same, the exchange rate profit makes it appear that the foreign stock returns 10%. The capita appreciation is due to foreign exchange rather than equity appreciation. Nevertheless, this can be profitable. Indeed, many international portfolio managers are choosing countries not only based on predictions of equities but also based on mode forecasting foreign exchange rates.

4. Private Equity vs. Public Equity

One final way we'll discuss the difference in categories is the difference between a public company and private equity. Pul equity means that the ownership of the shares of the company are made available to the public. Anyone who can trade or exchange in which that stock trades is eligible to purchase shares of stock, so anyone who has sufficient funds to pay for becomes an owner. Therefore, when a company is about to issue shares of stock, it is known as "*going public*". Compare with private equity. Sometimes, a company still wants to issue shares of stock but does not wish to do so through a publi offering. There are different reasons why a company may choose private funding rather than public. Briefly, one advantage using private equity is the reduced number of regulatory requirements, which are easier when dealing with private equity investors since they tend to be high-net-worth individuals or businesses. Furthermore, if the number of investors is small, may be easier to control the ongoing management of the company. Ultimately, there could be a subsequent offering that i public that will allow the private equity investors an even larger return by getting in "on the ground floor" of the company.

Remember that these choices are made not just for investors but from the company's perspective. A company wishing to public has probably already brought the company to a state where it can maximize the share value for the company's four For example, on the day that Mark Zuckerberg issued stock for Facebook (now Meta), his wealth increased by billions of c Because those stocks are now traded on a public exchange, it was very easy to evaluate his net worth. However, some companies, such as The Carlyle Group, a global investment firm, remain private investments. These companies need not v that an activist investor or other company could buy their shares because the shares are not offered on exchanges. Ultima private equity ownership gives investors control of the company without the worries of quarterly earnings, potential buyou other issues that affect publicly traded companies. In the long term, both are viable ways of raising funds for the original company.

5. Bitcoin

Now that we have seen different categories of stock, let us turn to different categories of cryptocurrencies. Asking an inve they like cryptocurrencies is like asking a music fan if they like jazz. For our music example, this question is broad becaus are many different types of jazz. It is certainly possible to like some forms of jazz and not others. Similarly, it is possible to some types of stocks (blue chips) and not others. For example, suppose you don't like penny stocks, which are valued at p with the potential to earn big returns but are also accompanied by high amounts of risk. These companies can make well thousand-percent return but can also lose 100% of the original investment.

Likewise, there are many kinds of cryptocurrencies. The oldest and most popular example remains Bitcoin. As of this writi Bitcoin is about $800 billion dollars or 40% of the approximately $2 trillion asset class of cryptocurrencies. Bitcoin's bigge contender is Ethereum, the second largest cryptocurrency. In addition to these two, there are many other cryptocurrencies can choose from, just like how there is a wide selection of stocks you can choose from. Unlike stocks, however, there arer many classifications of cryptocurrencies, like domestic and foreign, because cryptocurrencies transcend borders. Cryptocurrencies live in the global economy, in a digital world that literally operates 24 hours a day, 7 days a week. The Bit exchange never closes. One way to categorize cryptocurrencies is whether they are coins or tokens. Whereas coins are bu directly into the blockchain, tokens are built on top of an existing blockchain. Bitcoin and Ethereum are examples of coins tokens, respectively. Beyond this, there are thousands of other cryptocurrencies to choose from and trade. Certain feature also help to distinguish one cryptocurrency from another.

6. Bitcoin vs. Stocks

So, let's compare Bitcoin to stocks. Stocks derive an intrinsic value because they grant ownership to companies that provi goods and services. Cryptocurrencies, as their name suggests, are effectively a currency that has implied value through th mutual trust of a currency. To better understand cryptocurrencies in the economy, it helps to understand the role of money required reading *Money and Debt: The Public Role of Banks* by Stellinga et al. describes three criteria for currency and foc on the importance of trust that must remain within the currency system. In particular, be sure to read Box 1.2 Is Bitcoin mo Cryptocurrencies do not fulfill these three criteria even though they seem to have a degree of trust. Virtual Bitcoins are nea impossible, if not actually impossible, to counterfeit due to the extreme protocols by which a Bitcoin is created. Unfortuna though, it is possible to steal Bitcoin by accessing digital accounts. The fact remains that many cryptocurrencies are seen paradox by classical investors. Read section 2 of Kjærland's "An Analysis of Bitcoin's Price Dynamics" to understand the ty price dynamic Bitcoin has relative to stocks and other asset classes. Think about the "Greater Fool Theory," which sugges the only way to make money is for someone else to be convinced that it is worth more than you paid and give you a premi it.

7. Bitcoin vs. Other Alternative Investments

So how does Bitcoin compare to other alternative investments? Alternative investments include those investments that ar outside the mainstream of equities, fixed income, and other traditional asset classes. In fact, we've already discussed one example of an alternative investment: private equity. Private equity firms invest in different companies and hold them for l periods of time with the intention of building companies up, without worrying about the whims of missed quarterly earnin the distraction of public stockholders. They are intended for high-net-worth individuals who can afford to not have dividen income or to even suffer capital depreciation without sacrificing their quality of life. Other examples of alternative asset cl include real estate, such as homes and undeveloped land. We will be discussing the residential and commercial real estat markets in a subsequent module.

Another example of alternative assets is commodities. Commodities are an asset class that represents different products ranging from agricultural, financial, or natural resources. Indeed, The Commodity Futures Trading Commission defines agricultural goods (except onions due to a law that banned trading futures on onions!) and physical goods, such as precic metals like gold or aluminum; natural resources, such as energy, oil, or electricity; and financial instruments, such as a cur or interest rate. That list includes a lot of items! We'll visit commodities throughout the program. One of the interesting fea of commodities is the relationship of their supply and demand, as extreme weather, geopolitical risk, overproduction, panc and other disruptions can easily rattle the supply/demand dynamics and affect pricing worldwide.

Our last example of alternative assets is art. At first, it's hard to see art as an investment when you think about the bonds, and other securities whose markets dominate the financial news discussing default, management changes, earnings announcements, and product launches. Many investments provide periodic cash flows. Nevertheless, art can be an intrigu asset class for its ability to find "value artists" who become very valuable over time. Read the one-page abstract from Leve you will discover some interesting findings about the performance of art as an investment and how it compares to stocks

8. Bitcoin vs. Other Cryptocurrencies

Bitcoin is the most popular cryptocurrency; it is the one that dominates the headlines. Bitcoin moves typically make headl financial news daily. Let's take a look at the demographics of Bitcoin. According to a survey reported on by CNBC journalis best-selling author Robert Frank, most millennial millionaires use cryptocurrencies to store their wealth. Furthermore, 83% millennial millionaires own cryptocurrencies. Therefore, Bitcoin remains an exciting investment for many. It's as if you tool investing and fast-forwarded the movements, like in time-lapsed photography, so that a month's worth of volatility could h in a day. The size of Bitcoin's moves dwarfs virtually every type of stock. Combined with the fact that the Bitcoin market is more than three times the number of hours that the stock market is and that Bitcoin represents worldwide trust in a curre with accessibility, Bitcoin trading is like a 24-hour reality show with a worldwide audience. It is estimated that mentions of Bitcoin on social media appear approximately 20 times a minute (Bulao). Other cryptocurrencies simply don't have the bu market share. Since Bitcoin launched 15 years ago, there have been more than 15,000 cryptocurrencies, of which about 1(remain active. Ironically, one of those, Dogecoin, began as a joke with the image of the founder's dog on the front, only to become a serious cryptocurrency with $18 billion market capitalization.

One thing cryptocurrencies have in common with stocks is that there are low barriers to entry. The technology of a cryptocurrency is a variation on the blockchain technology. This is a relatively low barrier to entry and explains why there a 10,000 different cryptocurrencies that trade on exchanges. The same is true for some companies. It may not be very diffic penetrate the "advantage" the company has.

Let's look at an example of how it is possible to encroach on a company's competitive edge. Consider Peloton, for exampl Peloton was the first company to produce a bicycle that you could purchase for in-home use that had a smart tablet—a proprietary screen on which riders could attend live (or recorded) classes virtually in the comfort and convenience of their homes. Peloton tapped into a high-end market. For those who are unable to head to a gym or health facility to take a grou Peloton offered equipment and classes with convenience, motivation, and accountability. During the pandemic, Peloton's achieved new highs, largely due to the fact that people were required to socially distance and the opportunity to attend an person class was discouraged, difficult, or dangerous. Since Peloton's launch, there have been other entrants that produce similar apps on smartphones (like Zwift), similar bikes (like Echelon), or the combination of bike and software that have competed with Peloton's model. These new entrants provided lower upfront costs and lower subscription prices to get essentially the same workout. Potential Peloton customers sought out these competitors. As a result, Peloton slashed its That was not enough to maintain sales. Peloton was unable to attract new customers. As of this writing, Peloton has ceas production of their bikes. Perhaps the market is already saturated with its competitors and other products being offered. I the prices will drop, and new competitors will enter the market.

With such a company, it is difficult because the competitive edge is not something that is easily protected. In equities, this competitive edge is referred to as the competitive moat. A moat is a body of water that surrounded a castle in medieval ti prevent attackers from invading the inhabitants of the castle. Likewise, today's successful large-cap companies tend to ha moats that provide some protection, which new entrants cannot easily breach to enter the market. As a technological exa consider Verizon. To provide fiber optic service around different regions, there must be physical towers to set up. This is n easy thing to do, what with the purchase of land, the implementation of the technology, and the servicing. It is certainly pc to do, but there are very high barriers to entry, including high costs, accessibility to land, and infrastructure.

We discuss these issues here because the barrier to entry in cryptocurrencies is effectively the blockchain technology. So cryptocurrencies did very well because they were the first to enter the market, and there is no telling if that success will cc as the expectation is that new cryptocurrencies will continue to pour into the marketplace. With thousands of cryptocurre can some of those displace the market share that Bitcoin has? Time will tell.

In short, cryptocurrencies are a very difficult asset class to trade, in the sense that the volume of data, the newness of the market, and the lack of the traditional methods of valuation that apply for equities or fixed income simply make it a difficu class to model. How can we effectively decide if we should invest in stocks, Bitcoin, or any asset? Let's go back to statisti basics, the topic of our next section.

9. Conclusion

In this lesson, we addressed different ways of categorizing stocks and compared cryptocurrencies to each other in additio alternative investments. In the next lesson, it's time to get quantitative!

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